Decomposing Chinese Characters: LaTeX Typesetting with `zitex`

Wen G. Gong
wen.gong.research@gmail.com*

Abstract

We write the decomposition of a Chinese character in a few lines of data, and `zitex` turns it into a tree, a nested Zi-Matrix, a one-line formula and a typeset equation. This file is both a short paper and a guide: each block of markup is followed by a note on its syntax, and the features it shows. The characters here are 器藻佛.

1 Introduction

A character can be read as the product of a reaction: its parts go in, and the character comes out [1]. Writing that by hand in every paper is slow, and the layout of the parts (left–right, top–bottom, enclosing) is lost. With `zitex` the same thing in the middle of a sentence is 口 + 口 + 犬 + 口 + 口 → **器** (qì, device), and the layout is kept in the matrix of Figure 1.

Syntax. `\ziformula{字}` prints *parts → character (pinyin, gloss)* on one line, so it works inside a sentence, a caption or a table cell. It is the only style that is not a block.

The data. A character is a short YAML entry: its shape in prefix form, plus the position of each part in its level. The positions are the eleven Zi-Matrix codes `L R M U D Mi Mo LU LD RU RD`, where `M` is the centre of gravity and a Kangxi radical is peripheral. The characters of this paper come from a database, with `zitex extract`; see Table 1.

Syntax. The long form `\zi[style=formula]{器}` and the short form `\ziformula{器}` are the same. Every style has a shortcut: `\zitree`, `\zimatrix`, `\ziformula`, `\zimath`.

2 Four views of one decomposition

2.1 Equation: `\zimath`

The equation view is the most compact. Pinyin sits above each part, and the gloss below.

$$\underbrace{\overset{\text{kǒu}}{口}}_{\text{mouth}} + \underbrace{\overset{\text{kǒu}}{口}}_{\text{mouth}} + \underbrace{\overset{\text{quǎn}}{犬}}_{\text{dog}} + \underbrace{\overset{\text{kǒu}}{口}}_{\text{mouth}} + \underbrace{\overset{\text{kǒu}}{口}}_{\text{mouth}} = \underbrace{\overset{\text{qì}}{器}}_{\text{device}}$$

Syntax. `\zimath{字}` is a display equation made only of `\text`, `\overset`, `\underbrace` and `\xrightarrow`, so the same snippet works in MathJax. A part that has parts of its own is written as a bracketed group. The `lang` key picks the gloss language, as in `\zimath[lang=fr]{器}`, and needs that language in the lexicon.

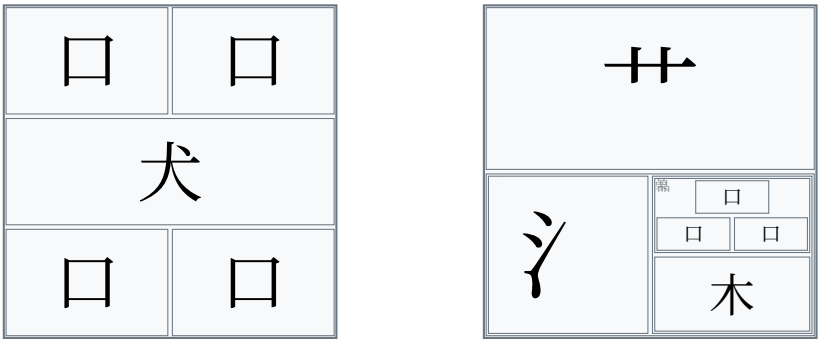


Figure 1: Zi-Matrices of 器 (left) and 藻 (right).

2.2 Matrix: `\zimatrix`

The matrix keeps the *geometry*: parts are boxes placed where the position codes say.

Syntax. `\zimatrix{字}` draws with TikZ and is an ordinary box, so it goes in a `figure`, a table cell, or a `\resizebox`. To change its size, scale the box: `\resizebox{0.8\linewidth}{!}{\zimatrix{器}}`.

2.3 Tree: `\zitree`

The tree shows how deep a decomposition goes.

Syntax. A tree can be wide, so Figure 2 is wrapped in `\resizebox`. The wrapper only scales, the snippet itself is unchanged. `\zitree` accepts the same keys as the others: `style`, `lang`, `reading`.

2.4 Choosing the style as an option

All four commands are shortcuts for one: `\zi[style=...]`.

Command	Output for 器	Use it for
`\zimath{器}`	an equation (display)	the main text, one idea per equation
`\zimatrix{器}`	nested boxes	figures, showing the layout
`\zitree{器}`	a tree	figures, showing the depth
`\ziformula{器}`	口 + 口 + 犬 + 口 + 口 → **器** (qì, device)	a sentence, a caption, a table

3 A small gallery

The characters 器藻佛 as a matrix and as a formula:

Syntax. Block styles work in a `tabular` as long as the column is wide enough, which is why the matrices are scaled with `\resizebox`. `\ref`, `\label` and `\caption` behave as for any figure or table, so a paper can refer to Table 1, Figure 1 and Figure 2 in the usual way.

4 Features you will want in a real paper

4.1 Language and reading

Equations and formulas print glosses, so they depend on a language. Set a default for the document or override it in one call:

*Every figure, equation and formula below comes from `\zi`-commands. The text around them is ordinary LaTeX.

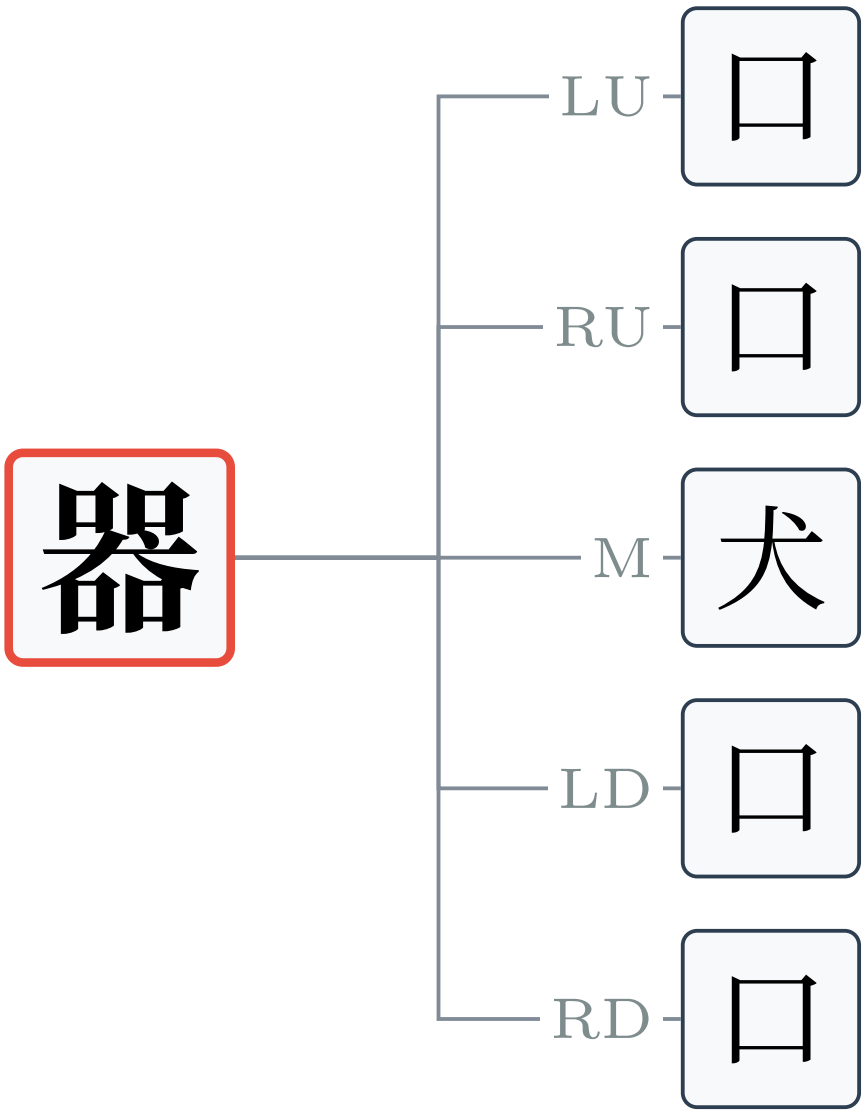


Figure 2: The decomposition tree of 器. Leaves are the parts that are not decomposed further.

```
\zitexsetlang{fr}                % glosses in French from here on
\zimath[lang=en]{器}                 % one call in English
\zitexsetreading{dictionary}     % use a named reading
\zimath[reading=author]{器}          % one call with another reading
```

Syntax. A *reading* is a named interpretation of the roles of the parts (semantic, phonetic, force), so two schools of thought can sit in the same data and a paper can choose. In the file system they appear as `<char>-<style>[-<lang>][@reading].tex`. These lines are not run here, because the lexicon of this folder has English only and no readings.

4.2 Where the snippets are

```
\zitexsetpath{zitex-out}         % the default: a folder next to main.tex
```

Syntax. To share one folder between several papers, point every paper to it, and run `zitex extract -o` on that folder.

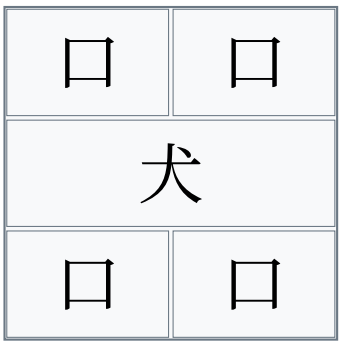
4.3 Fonts

The `zitex` package has a font fallback: when the primary font has no glyph for a part (rare characters, ancient components), a fallback font, an SVG image or an alias is used, as set in `fonts.yaml`. A part that nothing can draw prints a boxed code point such as `U+XXXX` and is listed by `zitex fonts`.

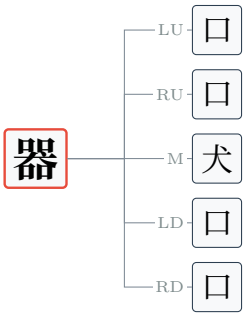
5 Every extracted character

器 (qì, device; tool; utensil)

$$\underbrace{口}_{\text{mouth}}{}^{\text{kǒu}} + \underbrace{口}_{\text{mouth}}{}^{\text{kǒu}} + \underbrace{犬}_{\text{dog}}{}^{\text{quǎn}} + \underbrace{口}_{\text{mouth}}{}^{\text{kǒu}} + \underbrace{口}_{\text{mouth}}{}^{\text{kǒu}} = \underbrace{器}_{\text{device}}{}^{\text{qì}}$$

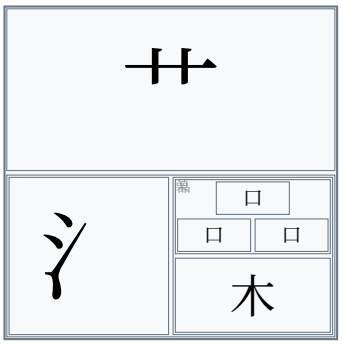


口 + 口 + 犬 + 口 + 口 → **器** (qì, device)

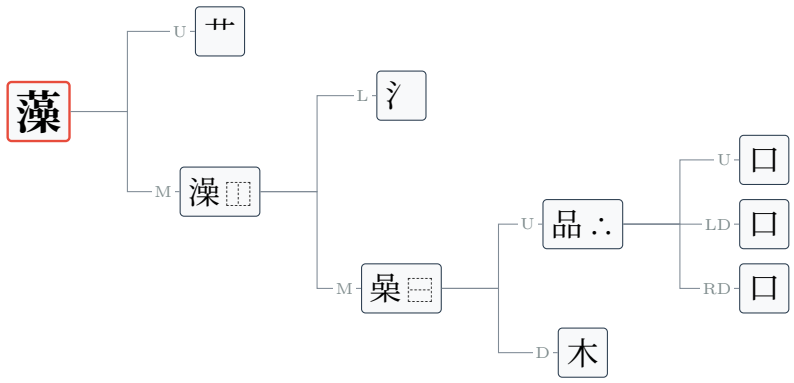


藻 (zǎo, aquatic grasses; elegant)

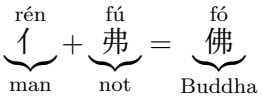
$$\underbrace{艹}_{\text{grass}}{}^{\text{cǎo}} + \underbrace{澡}_{\text{bath}}{}^{\text{zǎo}} = \underbrace{藻}_{\text{aquatic grasses}}{}^{\text{zǎo}}$$

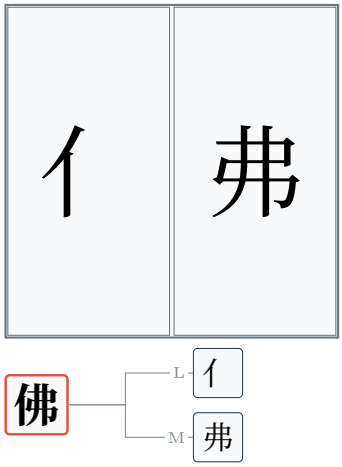


艹 + 澡 → **藻** (zǎo, aquatic grasses)



佛 (fó, Buddha; seemingly)





亻 + 弗 → **佛** (fó, Buddha)

6 Reproducing this paper

```
zitex setup   -o work/            # zitex.sty, fonts.yaml, glyphs/
zitex extract -c 器,藻,佛 -o work/ --annotation demo
cd work && xelatex main.tex       # the PDF
```

Syntax. Without `--annotation demo`, `extract` writes a minimal `main.tex` (every style for every character), and with it, this paper. `extract` writes this file again each time it runs. To keep a version you edited, copy it to another folder or name. To add a character, add it to the `-c` list and run `extract` again. If a character is missing, the red mark and the log line `Missing zitex-out/...` say which.

Summary of the syntax.

Markup	Meaning
`\usepackage{zitex}`	load the package (needs XeLaTeX)
`\zi[style=S, lang=L, reading=R]{字}`	general form; `S` is one of tree, matrix, formula, math
`\zitree`, `\zimatrix`, `\ziformula`, `\zimath`	shortcuts for the four styles
`\zitexsetlang{L}`, `\zitexsetreading{R}`	document-wide defaults
`\zitexsetpath{dir}`	where the snippets are (default `zitex-out`)

Acknowledgements

The design of `zitex` began as a conversation with Google Gemini, and the software, its tests and this document were written with the help of Anthropic Claude (Claude Code). The decompositions, the decisions about notation and the review of the results are the author's.

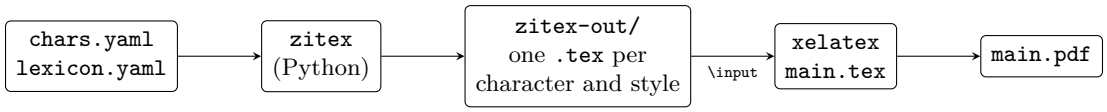
References

[1] Wen G. Gong. *A New Exploration into Chinese Characters: from Simplification to Deeper Understanding*. arXiv:2502.19428, 2025. `https://arxiv.org/abs/2502.19428`

A How LaTeX is extended: the workflow

`zitex` adds Chinese character decompositions to LaTeX without adding a large LaTeX program. **Python does the thinking, LaTeX does the typesetting.** Python reads the decompositions, works out the layout and writes small LaTeX files. A thin package, `zitex.sty`, pulls those files into the document.

A.1 The two halves



Half	Does	Written in
`zitex` command	parse a shape, check positions, choose a layout, compute coordinates, look up pinyin and glosses, resolve fonts, write LaTeX	Python
`zitex.sty`	load the packages the snippets need, define the `\zi` commands, find and `\input` the right snippet	LaTeX, about 60 lines

The document never contains a decomposition. It contains `\zimath{字}`, and LaTeX finds the snippet that Python wrote for that character.

A.2 Why not do it all in LaTeX macros?

- **Parsing is awkward in TeX.** A shape string is a nested prefix expression with fixed arities, named groups and several kinds of operator. In Python it is a short recursive function.
- **Layout needs arithmetic and tables.** The matrix is laid out recursively, and the flat patterns need a small solver.
- **Data does not belong in a document.** Decompositions, roles, glosses and fonts live in YAML files that people review, test and check (`zitex check`). LaTeX never sees them.

So the LaTeX side is kept simple on purpose: it knows only file names.

A.3 What `zitex.sty` does

1. **Loads what the snippets need:** `fontspec`, `xeCJK`, `amsmath`, `graphicx`, `tikz` and `xkeyval`. This is why `zitex` needs XeLaTeX.
2. **Sets the Chinese font** (`\setCJKmainfont`) and declares the operators ∴∵⁝ part of the `CJK` class with `\xeCJKDeclareCharClass`, so that `xeCJK` typesets them in the CJK font. Without it they print as nothing.
3. **Defines the commands** `\zi[style=, lang=, reading=]`, the four shortcuts, and `\zitexsetlang`, `\zitexsetreading`, `\zitexsetpath`.
4. **Includes the snippet, or says it is missing:** if the file exists, `\input` pastes it in. If not, LaTeX prints a warning and shows a red mark, so a missing snippet never stops a long build.
5. **Avoids one trap:** an empty macro made with `\newcommand` is not `\ifx`-equal to one made with `\def` (the first is `\long`), so the package uses `\def` for the reading.

A.4 The file name is the contract

Python writes the snippets and the package reads them, and the only interface between them is the file name:

```
zitex-out/<char>-<style>[-<lang>][@<reading>].tex
\zimath[lang=fr, reading=dictionary]{虹}   ->   zitex-out/虹-math-fr@dictionary.tex
```

The language is part of the name only for the styles that print glosses (`math`, `formula`), and `@reading` only when a reading was asked for. The rule exists twice, in Python (`snippet_name`) and in the package (`\zitex@name`), and a test fails if they differ.

A.5 What a snippet is

Python writes one self-contained file per character and style.

Style	Snippet
`math`	plain LaTeX math (`\overset` for pinyin, `\underbrace` for the gloss, `\xrightarrow` for the reason), which MathJax and KaTeX also accept
`formula`	one line of text: parts, an arrow, the character, its pinyin and gloss
`matrix`	TikZ, with rectangle coordinates that Python computed, and a fill colour from the role of each part
`tree`	TikZ, one node per component at computed coordinates; Python does the layout, not a TikZ tree library

Colours are defined inside every snippet, so it works when `\input` into any document. A snippet that needs a special font carries its own helper commands, declared with `\providecommand` so that they never clash with the package.

A.6 Fonts and glyphs that are not in the font

A component may be missing from the main font: a rare character, or a component with no Unicode code point. Python checks each component against the font files, and writes one of four forms:

Situation	What Python writes
the font has the glyph	the character itself
a fallback font has it	a group that switches the CJK font for that one character
it is drawn as an image	the image, included at text height
nothing can draw it	a boxed label such as `U+20009`, and a warning in the build output

A.7 How the pieces find each other

1. `zitex extract` (or `zitex render --scan main.tex`) reads the paper as text, notes the character, style, language and reading of each `\zi` command, and renders exactly those snippets into `zitex-out/`. It follows `\zitexsetlang` and `\zitexsetreading` in order, as the package does, and it skips comments and verbatim listings.

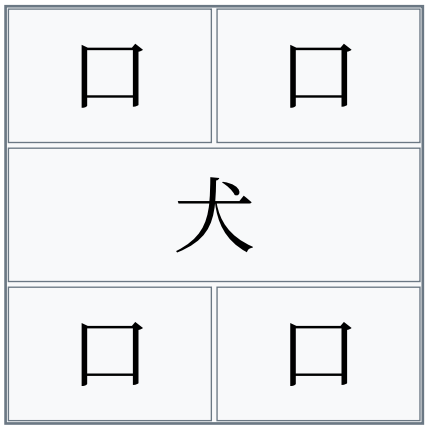
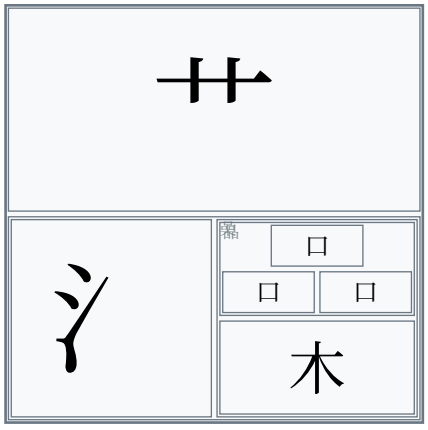
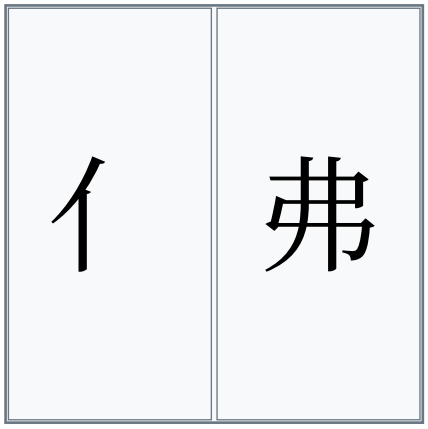
2. `xelatex main.tex` loads `zitex.sty`. In a folder made by `zitex setup`, the package sits next to `main.tex`, so nothing is installed anywhere else. `zitex install-sty` installs it for all documents.

3. At each `\zi` command, the package builds the file name and `\inputs` `zitex-out/<name>.tex`. The folder is relative to where `xelatex` runs, and `\zitexsetpath` changes it.

A.8 Readings

Which part of a character carries the meaning and which the sound is an interpretation, and a character can store several. The reading chosen only changes the colouring and the arrow label, and never the structure. It reaches Python through the file name, so the LaTeX side needs no new logic.

Table 1: One row per character. Cells hold `\zi` output directly.

	Matrix	Formula
器		口 + 口 + 犬 + 口 + 口 → **器** (qì, device)
藻		艹 + 澡 → **藻** (zǎo, aquatic grasses)
佛		亻 + 弗 → **佛** (fó, Buddha)